Nguyen Quoc Huy (Leo)

AI Engineer | Medical AI | Production ML Systems | LLM Post-Training & Evaluation

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Summary

AI engineer building Vietnamese medical AI end to end: clinical and synthetic data, post-training, evaluation and the production systems around them. Owner of a real-time computer-vision and speech proctoring service, taken from frame loss under load to stable multi-hour exams; started the company's LLM evaluation platform; built a large Vietnamese speech corpus and the ASR release benchmark. Works closely with product owners, QC and software engineers. Research background in medical time series and multi-agent VQA, with an AACR-presented paper.

Experience

AI Engineer Sep 2025 – Present

Trivita AI — healthcare and legal-tech AI Vietnam

Real-time AI exam proctoring — service owner since Jul 2026; FastAPI, Triton, LiveKit, Kafka, Kubernetes

- Took ownership of the proctoring service (face verification, gaze, pose, object, voice and keystroke signals) and improved its throughput **several-fold** while cutting per-frame latency.
- Load-tested it at the target concurrency for a full exam length with **no pod restarts and near-zero frame loss**; sized production autoscaling with headroom over the required frame rate.
- Root-caused frame loss in a live exam that happened while CPU sat mostly idle: dynamic input shapes made two detectors orders of magnitude slower, and Triton dynamic batching was far slower than batch-1. Fixed-shape inputs turned frame drops from the majority of frames into a small minority and cut cold-start time several-fold.
- Built a labelled video benchmark and made it a fast CI merge gate; recalibrated alert thresholds until every benchmark clip raised the correct alert level, and fixed recurring false alerts that blocked honest candidates.
- Built voice anti-cheat end to end (merged Sep 2026): LiveKit track egress → Kafka → VAD-gated speaker verification against enrolled voiceprints, with enrolment UI and a kill switch, at low bandwidth and CPU cost per candidate.
- Added Sentry, Prometheus/Grafana and structured logging; fixed DB connection-pool exhaustion and automatic video recovery after network loss; introduced decision records and trunk-based GitOps deploys with ArgoCD; on call for the live pilot exam.

Medical AI: MedVita, clinical data and post-training — vLLM, SFT/DPO, LoRA/QLoRA, DVC, MinIO, Label Studio

- Built **MedVita**, a Vietnamese medical history-taking assistant: extraction and normalisation of symptoms, labs, diseases and drugs to **ICD-10 VN**, negation and family/past-history context detection, SOCRATES-structured history-taking per department, tiered safety guardrails; integrated with a central hospital's HIS/EMR with idempotent writes and audit logs.
- Built the real-world medical data pipeline: EMR/HIS and multi-dialect audio ingestion, PII/PHI redaction before any prompt, Label Studio golden batches, versioning on DVC + MinIO + Hugging Face Hub, MinHash dedup and contamination checks.
- Generated a **large synthetic patient-persona set** matched to national census (GSO) distributions, crossed with clinical scenarios and risk profiles, batched through vLLM with prefix caching and CPU offload.
- Co-built the post-training loop (eval → targeted data generation → SFT/DPO with LoRA/QLoRA) on an on-prem **multi-node H100 cluster** with day-scale iteration cycles; served models with vLLM tensor parallelism.
- Ran a prompt/harness ablation for a clinical-note agent on real pediatric visits with blind LLM judging: spec redesign raised an open-weight model's score substantially and showed the golden reference itself fell short of the rubric.

LLM evaluation platform — FastAPI, Celery/Redis, Postgres, DeepEval, DeepTeam, Langfuse, Next.js

- Started the company's self-hosted LLM evaluation platform and built its first working version single-handedly (API, async pipeline, CLI, Docker, test suite); it is now developed by several engineers. Set the team rule of *eval before training*.
- Designed medical G-Eval and LLM-as-judge metrics (summary hallucination, empathy, emotional intelligence), conversational eval and red-teaming (jailbreak, prompt injection, PII leakage); added MedHALT and lm-eval runners with a self-hosted judge on H100; built the SOCRATES history-taking eval across multiple models and clinical scenarios.
- Migrated the medical chatbot's tracing to Langfuse v4 (fixing fragmented traces and connection-pool exhaustion) and built a patient-simulation harness for conversational testing.

Vietnamese medical speech (ASR / TTS) — data, evaluation and serving

- Built a **multi-thousand-hour** Vietnamese speech corpus from dozens of sources with a streaming build–upload–prune pipeline and licence gate, cutting storage substantially.
- Designed a stratified benchmark of the production speech-to-text API across clinical, telephony, broadcast and general audio with bootstrap confidence intervals: it showed server-side denoising made WER worse on every tier, and that WebM/Opus is far smaller than WAV at negligible accuracy cost.
- Root-caused a text-to-speech regression to an API layer silently dropping the reference transcript; the fix and a guard brought production round-trip WER back to **near zero**.

- Supported coordination of the speech-annotation QC workflow (golden batches, inter-annotator agreement; authored most of the gold diarization annotations) and built the healthcare ASR release gate with a robustness suite.

Cross-team work

- Supported coordination and quality assurance with product owners, QC and software engineers: turned product requirements into acceptance criteria, triaged QC-reported bugs, and flagged product gaps to other teams (exam UX, consent for face data, multi-monitor setups, appeals flow).
- Mentored an intern to independent delivery on every handed-off task (diarization pipeline, Langfuse-integrated eval UI, eval landscape survey, ASR/TTS labelling); review code across the AI team.
- Supported the speech-team handover; ran an internal seminar on agent-assisted engineering. Technical contributor to company patent filings and to hospital and government demos that supported a national-level research project.

Research Before Industry

AI Researcher, *AIMA* Jun 2024 – 2025

- Medical time-series forecasting and anomaly detection with doctors and clinical experts.

Multi-Agent VQA Research Team Lead, *VLAI (remote)* 2023 – 2024

- Research on multi-agent visual question answering: experiment design, task coordination, progress reporting.

Time-Series Research Team Lead, *AI VIET NAM, AI Residency program (remote)* 2022 – 2023

- Deep learning models for medical time-series data; mentored junior members and set milestones.

Project Lead, Mental Health Support System, *Vaasa University of Applied Sciences*

- RAG-based (LlamaIndex) mental-health support system built with an international student team; awarded a perfect score by the university evaluation committee.

AI Intern, *DzStar Digitals* 2023

- Agent and automation prototypes integrated with Google Cloud and AWS.

Publication & Community

- Nguyen Huy, et al. *Convolutional neural network-based artificial intelligence for immune subtype classification in pan-cancer.* Presented at the American Association for Cancer Research (AACR) Annual Meeting.
- Co-founder, **AIO Learners Hub** — AI learning community on Discord.

Projects & Competitions

- **VLSP 2026 shared tasks** (Vietnamese speaker verification, expressive TTS, speech-quality assessment): an ensemble speech-quality predictor (HuBERT, XLS-R, WavLM) that beats the best single model; best-of-N reranking for non-verbal TTS.
- **EEP** | github.com/NguyenHuy190303/EEP: open-source agent-skills plugin for Claude Code and Codex, with its own with/without-skill eval harness.
- **vllm-autotuner** | github.com/MooSun-Leo/vllm-autotuner: search for the fastest vLLM serving configuration on multi-GPU hosts.
- **Capybara** (founder, side project, 2025–): multi-agent learning platform with a content scraper and AI tutoring.

Skills

- **Clinical NLP & retrieval:** medical NER, concept normalisation (ICD-10 VN, RxNorm), negation/context detection, relation extraction, RAG and semantic retrieval (embeddings, FAISS), LlamaIndex, LangChain.
- **Post-training & eval:** SFT, DPO, LoRA/QLoRA, synthetic data generation, LLM-as-judge, G-Eval, DeepEval, DeepTeam, lm-eval-harness, Langfuse, W&B, red-teaming.
- **Training & serving:** PyTorch, TensorFlow, distributed training (DP/TP/PP, ZeRO, FSDP; Megatron-LM), vLLM, TGI, TensorRT-LLM, Triton Inference Server, ONNX Runtime; on-prem H100 cluster, GB200 NVL72 (partner access).
- **Vision & speech:** YOLO, InsightFace (SCRFD/ArcFace), OpenCV, speaker verification (CAM++), Silero VAD, ASR/TTS evaluation (WER, DER).
- **Systems:** Python, FastAPI, Celery, Redis, PostgreSQL, Kafka, LiveKit/WebRTC, DVC, MinIO, Label Studio, Docker, Kubernetes, Helm, ArgoCD, GitHub Actions, SkyPilot, Prometheus, Grafana, Sentry, TypeScript/Next.js, SQL, C.

Education & Certifications

Bachelor's degree, Information Technology 2021 – 2025

Vaasa University of Applied Sciences (VAMK), Finland

AIO 2024 — AI & Data Science program 2024 – 2025

AI VIET NAM — certificate in GenAI & LLMs, CV & NLP, DL & ML

CS50x, CS50 AI, Using Python for Research (HarvardX, 2024) • NASA Space Apps Challenge 2024 — Galactic Problem Solver
Languages: Vietnamese (native), English (C2), Swedish (A2).